

Course Code	Course Title	L	T	P	C
BCSE412L	Parallel Computing	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					

Time –Analysis of PRAM - Asymptotic Analysis of Parallel Programs - Other Scalability Metrics.			
Module:6	Parallel Programming		7 hours
Shared Memory Programming - Distributed Memory Programming– Distributed Shared Memory – Message Passing – Programming Using the Message Passing Paradigm – Group Communication – Heterogeneous computing systems – Case Study (RPC and Java RMI).			
Module:7	Parallel Algorithms		5hours
Matrix Multiplication - Sorting Algorithms - Graph Algorithms – Applications			
Module:8	Recent Trends		2 hours
Guest lectures from Industry and, Research and Development Organizations			
	Total Lecture hours:		45 hours
Text Book(s)			
1.	Ananth Grama, Anshul Gupta, George Karypis and Vipin Kumar, "Introduction to Parallel Computing", Pearson, 2nd Edition, 2015.		
2.	David Kirk, Wen-mei W. Hwu, Programming Massively Parallel Processors - A Hands-on Approach, Morgan Kaufmann, 3rd Edition, 2016.		
Reference Books			
1.	Michael J. Quinn, Parallel Computing: Theory and Practice, 2nd edition, McGraw Hill Education, India, 2017.		
2.	Ian Foster , Gerhard R. Joubert, Ludek Kucera, Wolfgang E. Nagel, Frans Peters, Parallel Computing: Technology Trends: Advances in Parallel Computing, IOS Press,2020.		
Mode of Evaluation: CAT / written assignment / Quiz / FAT			
Recommended by Board of Studies		12-05-2023	
Approved by Academic Council		No. 70	Date 24-06-2023